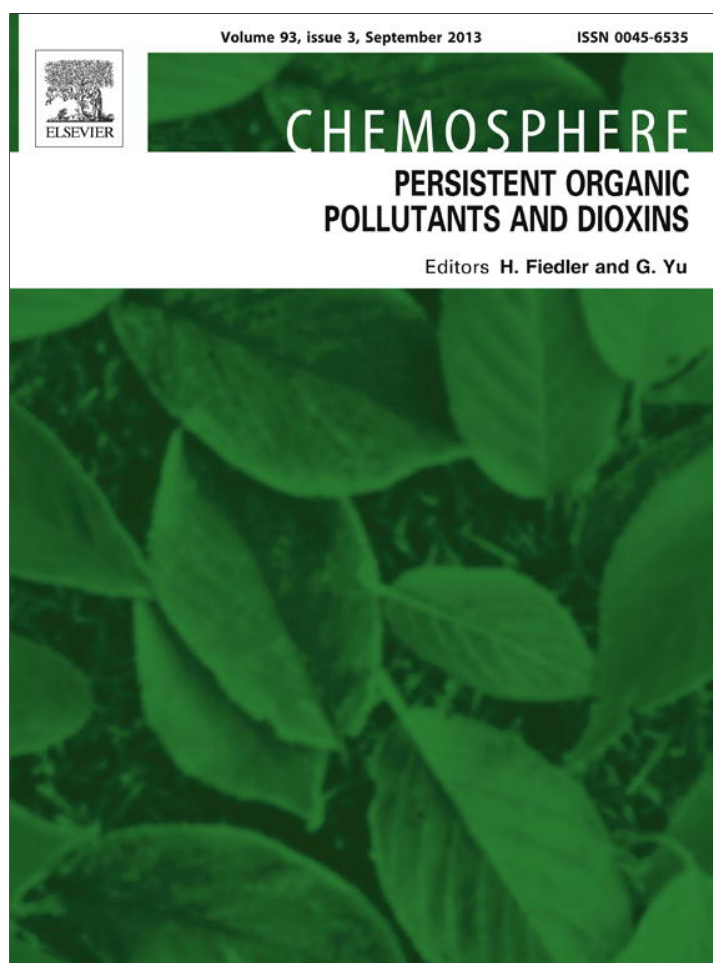




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A probabilistic model for the carry-over of PCDD/Fs from feed to growing pigs



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HIGHLIGHTS

- A pharmacokinetic model was combined with a probabilistic model to gain a decision tool for risk management.
- The German dioxin incident of winter 2010/2011 can be described by the tool.
- Clearing rates for the congener 1,2,3,4,7,8-HxCDD were determined from experimental data ([Hoogenboom et al., 2004](#)).
- The approach demonstrated here is a showcase how a risk assessment in the case of contaminated feeding can be performed.

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ABSTRACT

When food producing animals are contaminated with PCDD/F congeners, information on the contaminant's concentration in the bodies of the animals at time of slaughter is needed for risk management purposes. We have developed a mathematical model for the kinetics of PCDD/Fs in growing pigs in case of contaminated feed fed for a limited duration of time. This model allows the prediction of concentrations in body fat. It considers absorption fractions of PCDD/Fs, clearance by metabolism, dilution by growth and excretion through fecal fat. The model parameters were calibrated by fitting the model to experimental data. On the basis of this toxicokinetic model a probabilistic model has been constructed. The probabilistic model handles the parameters with appropriate probability distributions and Monte-Carlo simulation technique, providing for realistic situations with many animals and a range of contaminations and feeding intervals. We applied the new model to describe the German dioxin incident of winter 2010/2011 and discuss its viability as decision tool. The approach demonstrated here is a showcase how a risk assessment in the case of contaminated feeding can be performed.

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1. Introduction

Contaminations of animal feed for food producing animals with PCDD, PCDF and dioxin-like PCB have recurred irregularly, for instance 1999 in Belgium ([Van Larebeke et al., 2002](#)), 2008 in Ireland ([Heres et al., 2010](#)) and 2010/2011 in Germany ([Abraham et al., 2011](#)). As a result of the toxicological properties of PCDD, PCDF and dioxin-like PCB this is of public health concern ([WHO, 2010](#)). During the German dioxin incident of 2010/2011, a quantitative assessment was needed to study the kinetics of PCDD/Fs in pigs. Previous models looked at worst-case or specific scenarios ([Fries, 1996](#); [Hoogenboom et al., 2007](#)). To investigate all possible outcomes and scenarios, a more general model was needed. We constructed a probabilistic model by combining a toxicokinetic model with a Monte-Carlo simulation, where all important parameters were modeled using probability distributions. This

probabilistic model is capable of providing risk assessment in the case of future dioxin incidents. Beyond that, it can help to prevent gratuitous culling and to avert further damage to farmers.

The model is calibrated with the experimental data of [Hoogenboom et al. \(2004\)](#), who conducted a feeding experiment with growing pigs using diluted feed from the dioxin crisis in Belgium in 1999. Furthermore, we refer to the following studies: [Lenk \(2007\)](#), who accomplished a similar feeding experiment as [Hoogenboom et al. \(2004\)](#) with three different concentrations of a PCDD/F mixture; [Shen et al. \(2012\)](#), who repeated Lenk's experiment and constructed a prediction formula for the PCDD/F content in food; and [Spitaler et al. \(2005\)](#), who fed pigs with contaminated feed of different concentrations of a PCDD/F mixture and measured the PCDD/F concentrations in different tissues. Previous theoretical approaches with point estimates were made by [Fries \(1996\)](#), who used a kinetic one-compartment model without considering metabolic clearance, excretion and uncertainty, and [Hoogenboom et al. \(2007\)](#), who performed simulations of PCDD/F concentrations in growing pigs with a two-compartment

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model. Although these kinetic models differ from our model, they were used as an additional test for the validity of ours.

2. Materials and methods

In the German dioxin incident of winter 2010/2011 a peculiar mixture of PCDD/F congeners was observed in contaminated feed. The toxic equivalent (TEQ) level (Van den Berg et al., 1998) was mainly caused by two hexachlorodibenzo-p-dioxins (1,2,3,6,7,8- and 1,2,3,7,8,9-HxCDD). Both congeners also were present in the experiment of Hoogenboom et al. (2004), but their concentrations in body fat are too close to the limit of detection, or even below. In the experiment of Hoogenboom et al. (2004) the congener 1,2,3,4,7,8-HxCDD was present in a higher concentration, hence we use these data to calibrate our kinetic model of the pig. Under the assumption that the three HxCDDs behave similarly due to their similar chemical structure, the model can be used to simulate the German incident. Hoogenboom et al. (2004) used diluted feed from the Belgian dioxin crisis in 1999. During the experiment, growing pigs were fed for one week with contaminated fodder. Afterwards, they were kept under normal conditions up to an age of 6 months. At six time points after exposure, samples of back fat were taken by biopsy and analyzed. Lenk (2007) conducted a similar feeding experiment with three groups of pigs and three different concentrations of a PCDD/F mixture. We performed the same model-fit with their data as we did with the data of Hoogenboom et al. (2004).

2.1. Model description

The present model describes the kinetics of PCDD/F carry-over from feed to body fat of growing pigs. Generally, measured PCDD/F values are related to fat, concerning the highly lipophilic character of PCDD/Fs. Samples from different tissues of the same individual usually agree very well due to homogeneous distribution corresponding to fat content (Beck et al., 1994). Therefore, we use a pig model that consists of a single fat compartment of age dependent size. PCDD/F uptake and distribution in body fat is fast compared to characteristic times of PCDD/F elimination.

The total PCDD/F amount in the pig changes with PCDD/F intake with feed, metabolic clearance and excretion via faeces. The PCDD/F absorption fraction and metabolic clearance rate are estimated with this model from Hoogenboom et al. (2004) data. Since Hoogenboom et al. (2004) did not measure the body fat of the pigs, we use the values given in Pfeiffer et al. (1984). For the feed amounts we use values given in Staudacher (2010). The excreted amount of fat per day is assumed to be around 1 g (Guercioli et al., 2001).

The change of PCDD/F amounts in body fat is described as sum of incoming and outgoing amounts of PCDD/F (Eq. (1)). The uptake by feed is described as the product of the PCDD/F concentration in feed, the daily feed portion and a constant absorption fraction. The amount of metabolized PCDD/Fs is calculated as a product of the total amount of PCDD/F in the pig and the metabolic clearance rate. The excreted amount is calculated by the product of PCDD/F concentration in body fat and fat excretion rate.

$$\frac{d}{dt}A(t) = C_f(t) I(t) k_{abs} - A(t) k_{cle} - C(t) k_{exc} \quad (1)$$

The symbols, their meaning and units are listed in Table 1. The amount of PCDD/F in a pig at the beginning of the fattening period is assumed to be zero. In the final analysis, we want to describe the PCDD/F concentration $C(t)$ in body fat of pigs. Therefore, we need to introduce functions for the feed quantity $I(t)$ and body fat $F(t)$ into Eq. (1) (Appendix A).

Table 1

Symbols, meanings and units.

Symbol	Meaning	Unit
$W(t)$	Body weight	kg
$F(t)$	Body fat	kg
$I(t)$	Daily feed intake	kg d ⁻¹
$A(t)$	Total amount of PCDD/F in a pig	ng
$C(t)$	PCDD/F concentration in body fat	pg g ⁻¹
$A_f(t)$	PCDD/F amount in feed	ng
$C_f(t)$	PCDD/F concentration of feed dry matter	ng kg ⁻¹
k_{abs}	Absorption fraction	–
k_{cle}	Metabolic clearance rate	d ⁻¹
k_{exc}	Excretion rate via feces	g d ⁻¹

2.2. Growth functions and feed function

The body weight is described by a power function with offset (Eq. (2)) fitted to the data of Pfeiffer et al. (1984). As can be seen in Fig. 1 (left), body weight and body fat are highly correlated. We divided the experimental body fat by the experimental body weight and obtained the data of Fig. 1 (center), showing that the body fat fraction of pigs older than 75 d grows linearly with time. Using $F(t)/W(t) = f_1 t$ for the fat fraction, we obtain Eq. (3) for the body fat. The amount of feed as function of body weight (Staudacher, 2010) converges to a constant and can be described by an exponential saturation function (Eq. (4)).

$$W(t) = w_0 + w_1 t^{w_2} \quad (2)$$

$$F(t) = f_1 t W(t) \quad (3)$$

$$I(W(t)) = i_1 (1 - e^{-i_2 W(t)}) \quad (4)$$

The coefficients of Eqs. (2)–(4) are listed in Table 2. The data fitting was performed with the Levenberg–Marquardt nonlinear regression algorithm (Marquardt, 1963), using the GNU Octave-function `leasqr` (Eaton et al., 2009, <http://octave.sourceforge.net>).

Substituting Eqs. (2)–(4) into the differential equation for the PCDD/F concentration in body fat yields a differential equation (Eq. (A.5)), that contains 9 parameters ($w_0, w_1, w_2, f_1, i_1, i_2, k_{exc}, k_{abs}, k_{cle}$). Five of these parameters are estimated from the fit of growth and feed functions (Table 2), the birth weight was set to $w_0 = 1.3$ kg (Pfeiffer et al., 1984) and k_{exc} is assumed to be 1 g d⁻¹ (Guercioli et al., 2001). Therefore two unknown parameters remain, k_{abs} and k_{cle} .

The differential equation for the PCDD/F concentration $C(t)$, Eq. (A.5), is solved with a finite-difference method, using constant time steps of 1 d. We compared this method with the more sophisticated Dormand–Prince method (Dormand and Prince, 1980), implemented in the GNU Octave-function `ode45` (Eaton et al., 2009, <http://octave.sourceforge.net>), and found only minor differences between the methods. Due to the better computing performance of the finite-difference method, we went further with that method.

2.3. Calibration of the model

To calibrate the model, the metabolic clearance rate k_{cle} and the absorption fraction k_{abs} of the differential equation Eq. (1) were fitted simultaneously to the experimental data of Hoogenboom et al. (2004) (Fig. 2, left). The fit was performed with the Levenberg–Marquardt algorithm (Marquardt, 1963), using the GNU Octave-function `leasqr` (Eaton et al., 2009, <http://octave.sourceforge.net>). For the absorption fraction we obtained the fit result $k_{abs} = 0.456 \pm 0.013$ and for the metabolic clearance rate $k_{cle} = (3.8 \pm 1.6) \times 10^{-3}$ d⁻¹ (Table 2), corresponding to a half life of $t_h = 180$ d with a 95%-error-interval of [100, 1040] d.

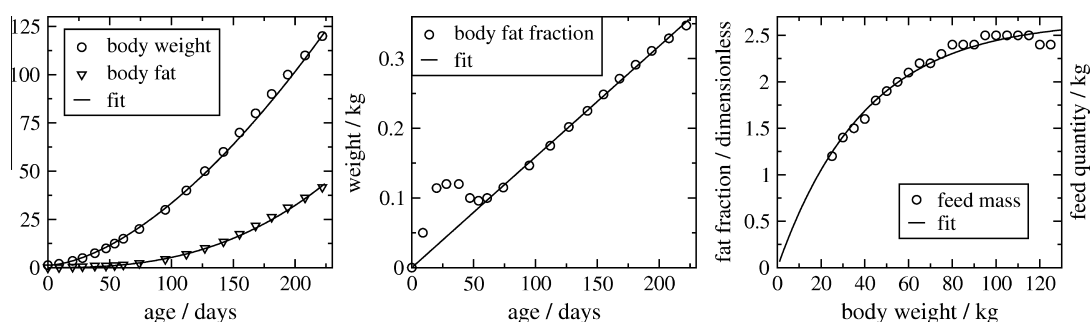


Fig. 1. (Left) Body weight and body fat as functions of age. Data from Pfeiffer et al. (1984), fitted to a power law with offset using the Levenberg–Marquardt algorithm. (Center) Body fat fraction calculated from data (Pfeiffer et al., 1984), fitted with a linear function. (Right) Feed quantity as a function of body weight. Data from Staudacher (2010), fitted with a sigmoidal function using Levenberg–Marquardt algorithm (Marquardt, 1963).

Table 2
Mean values μ , standard deviations σ and Pearson correlation coefficients ρ (p_1, p_2) of the parameters $p_1 = w_1, f_1, i_1$ and $p_2 = w_2, i_2$. Body weight $W(t) = w_0 + w_1 \cdot t^{w_2}$ with $w_0 = 1.3$ kg and body fat fraction $F(t)/W(t) = f_1 \cdot t$ fitted to the data of Pfeiffer et al. (1984). Feed quantity $I(W) = i_1 \cdot (1 - e^{-i_2 \cdot W})$, fitted to the data of Staudacher (2010). Differential equation Eq. (1) fitted to the data of Hoogenboom et al. (2004).

	p_1	μ	σ	p_2	μ	σ	$\rho(p_1, p_2)$
$W(t)$	w_1	0.0170	0.0018	w_2	1.64	0.02	−0.99
$F(t)$	f_1	1.589×10^{-3}	0.006×10^{-3}	–	–	–	–
$I(t)$	i_1	2.66	0.04	i_2	0.0252	0.0011	−0.90
Eq. (1)	k_{cle}	3.8×10^{-3}	1.6×10^{-3}	k_{abs}	0.456	0.013	+0.67

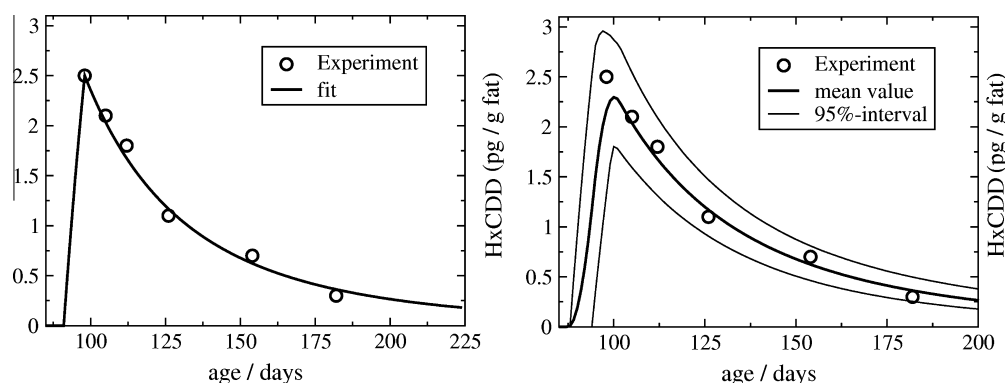


Fig. 2. PCDD concentration of congener 1,2,3,4,7,8-HxCDD measured by Hoogenboom et al. (2004) (circles). (Left) Resulting curve for the fit of metabolic clearance rates and absorption fractions k_{cle} and k_{abs} (straight line), Table 2. (Right) Plausibility check of the probabilistic model. Normal distributions for parameters $w_1, w_2, f_1, i_1, i_2, k_{abs}, k_{cle}$ from Table 2. Uniform distributions for parameters C_i, t_{leng}, t_{intv} (Table 3, scenario 3).

2.4. Probabilistic model

The concept of a probabilistic model is to incorporate parameter uncertainty and variability and to study their effect on the model results. The probabilistic model includes the kinetic model described above, embedded in a Monte-Carlo simulation. We used the software GNU Octave (Eaton et al., 2009) to build the probabilistic model. In this model, 10 parameters are drawn from probability distributions to consider their uncertainty given by respective data and model. These parameters are: w_1, w_2, f_1 – parameters of growth functions; i_1, i_2 – parameters of feed function; k_{cle}, k_{abs} – metabolic clearance rate and absorption fraction; t_{leng}, t_{intv} – length and limits of contamination interval; C_i – feed contamination level. t_{leng} is the time interval length of contaminated feeding, while t_{intv} gives the lower and upper time interval boundaries, between which the time interval varies.

The parameters C_i, t_{leng} and t_{intv} are modeled with uniform distributions, reflecting exposure scenarios of the pigs. An exposure scenario defines the feed contamination, the duration of contaminated feeding and the pig age at beginning and end of contaminated feeding. In this work we investigated three exposure scenarios (Table 3) describing: 1. The German incident of winter 2010/2011. 2. The

background contamination in Germany (Schwind et al., 2009). 3. The experimental setting of Hoogenboom et al. (2004).

The parameters $w_1, w_2, f_1, i_1, i_2, k_{cle}, k_{abs}$ are modeled with normal distributions using mean values μ and standard deviations σ from fit (Table 2). For simultaneous fitted parameters also Pearson correlation coefficients ρ are estimated during the Levenberg–Marquardt fit (Table 2). These correlation coefficients are used to construct correlated normal distributions for Monte-Carlo sampling (Appendix B). The parameters $(w_1, w_2), (i_1, i_2)$ and (k_{cle}, k_{abs}) are pairwise correlated. The correlation between body weight and body fat is taken into account by defining the body fat as function of body weight (Eq. (3)), while the correlation between body weight and feed quantity is included implicitly in the definition of the feed quantity as function of body weight (Eq. (4)). The birth weight w_0 is not modeled with a probability distribution because its variation is negligible.

The Monte-Carlo simulation proceeds as follows: The PCDD/F concentration as a function of time is calculated $N = 10000$ times. For each of the N Monte-Carlo runs, each parameter is chosen at random, according to its probability distribution. Therefore we obtain N time-dependent concentration curves $C_1(t), \dots, C_N(t)$. We are interested in the statistical properties of the N simulated

Table 3

Scenario-parameters. 1. German incident (Abraham et al., 2011), 2. Background Germany (Schwind et al., 2009), 3. Experiment (Hoogenboom et al., 2004).

Scenario	C_f	t_{feng} (weeks)	t_{intv} (weeks)
1. German incident 2010/11	[0.04, 0.47] ng TEQ kg ⁻¹	[8, 10]	[12, 32]
2. Background Germany	[0.018, 0.043] ng TEQ kg ⁻¹	20	[12, 32]
3. Hoogenboom (HxCDD)	[2.9, 3.5] ng kg ⁻¹	[0.86, 1.14]	[12.5, 14.5]

curves that can be achieved using a 95% simulation envelope around the mean value curve. Technically, the mean value curve is a combination of the mean values of N independent estimates at each time point. The curves for the 2.5% and 97.5% – quantile (forming the 95% envelope) are calculated analogously.

2.5. Sensitivity analysis

We have applied a sensitivity analysis to investigate the influence of input parameters on the outcome. This sensitivity analysis is based on the simulation approach described above. It investigates the correlation between result and input parameters. Technically, the seven normally distributed parameters are vectors of random numbers with N elements (N is the number of Monte Carlo steps). We analysed the influence of the parameters on the PCDD/F concentration at slaughter age. For this purpose we investigated Spearman rank correlations between each of the seven parameter vectors and the PCDD/F concentrations $C_1(t_{\text{end}}), \dots, C_N(t_{\text{end}})$ at $t_{\text{end}} = 220$ d. As usual, these rank correlation coefficients are shown as a horizontal bar plot.

3. Results

3.1. Plausibility check of the probabilistic model

Since the parameters of the simulation model were calibrated to the experimental data of Hoogenboom et al. (2004), also the probabilistic model must describe these data, if the probability distributions are chosen appropriately. In Fig. 2 (right) the parameters $w_1, w_2, f_1, i_1, i_2, k_{\text{abs}}, k_{\text{cle}}$ were sampled from normal distributions with values from Table 2. The three remaining parameters $C_f, t_{\text{feng}}, t_{\text{intv}}$ are sampled with uniform distributions (Table 3, scenario 3) describing the experimental conditions of the Hoogenboom experiment. In Fig. 2 (right) the mean value, as well as the 95% envelope curves are shown. The experimental values are well described by the mean-value curve, while the 95% envelope encloses them as intended.

3.2. Investigated scenarios

The three parameters $C_f, t_{\text{feng}}, t_{\text{intv}}$ have to be chosen either according to the experimental setting or according to the estimations for a dioxin incident. In Table 3, three scenarios are listed.

Scenario 1 describes the German dioxin incident. In 50 suspicious samples, concentrations between 0.039 and 0.468 ng TEQ kg⁻¹ feed were found (Abraham et al., 2011). Pigs were fed contaminated fodder at most 10 weeks in the fattening period of 20 weeks. In the simulation we used a feeding interval of 8–10 weeks. In Fig. 3 the mean values μ and 95% – uncertainty envelopes (composed of the 2.5% and 97.5% – quantile curves, $Q_{0.25}$ and $Q_{0.975}$) of the PCDD/F concentrations are shown. The $Q_{0.975}$ – curve is above the maximum food level (The European Commission, 2011) of 1.0 pg TEQ g⁻¹ fat (dotted line) for a short period. This is in accordance with the German incident, because only 3 of 124 samples were above the maximum level (Abraham et al., 2011), while the mean was 0.3 pg TEQ g⁻¹ fat and the minimum 0.1 pg TEQ g⁻¹ fat.

Scenario 2 is on the basis of a German PCDD/F background study (Schwind et al., 2009) with 206 samples of compound feed. The quantiles $Q_{0.25}$ and $Q_{0.75}$ were found to be 0.018 and 0.043 ng TEQ kg⁻¹ feed. For that scenario, the contaminated feeding was fixed to 20 weeks (complete fattening period from age of 12–32 weeks) with concentrations of that $[Q_{0.25}, Q_{0.75}]$ ng TEQ kg⁻¹ feed interval. The $Q_{0.975}$ – curve does not exceed a value of 0.12, i.e. is far below the maximum food level (Fig. 3).

Scenario 3 uses the parameters for Hoogenboom et al. (2004) experiment, considering uncertainty (Fig. 2, right).

Remark. The calculations were done for the concentration of HxCDD. The TEQ-values were calculated for comparison with the maximum food level. We used the fact that the total TEQ in the German incident was mainly caused by two HxCDD congeners (we assumed completely) and that all HxCDD congeners have the same TEF = 0.1. The unit ng TEQ kg⁻¹ refers to ng WHO-PCDD/F-TEQ kg⁻¹ (Van den Berg et al., 2006).

3.3. Sensitivity analysis

The seven parameters that describe feed and growth functions as well as absorption fraction and metabolic clearance rate ($w_1, w_2, f_1, i_1, i_2, k_{\text{abs}}, k_{\text{cle}}$) are sampled with normal distributions using their mean values and standard deviations from fit (Table 2). The three remaining parameters, PCDD/F concentration in feed as well as length and limits of feeding interval ($C_f, t_{\text{feng}}, t_{\text{intv}}$), are not included in the sensitivity analysis. The seven normally distributed parameters from the fit are not comparable with the three scenario-parameters in a reasonable sensitivity analysis, because the scenario parameters are relatively arbitrary and described by uniform distributions.

The Spearman correlation coefficients are displayed as horizontal bars (Fig. 4), showing that the resulting PCDD/F concentration is highly sensitive to the two parameters k_{cle} and k_{abs} . The sensitivity

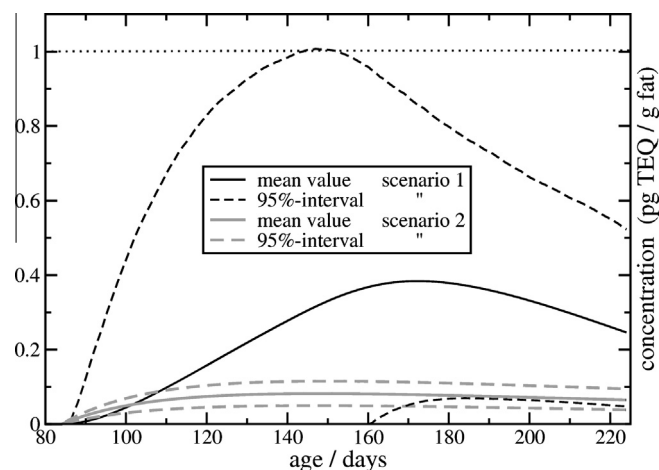


Fig. 3. PCDD/F concentrations of scenario 1 (German dioxin incident, black) and scenario 2 (German background, gray), with parameters $C_f, t_{\text{feng}}, t_{\text{cent}}$ from Table 3 and $w_1, w_2, f_1, i_1, i_2, k_{\text{abs}}, k_{\text{cle}}$ from Table 2. Maximum food level: 1 pg TEQ g⁻¹ fat.

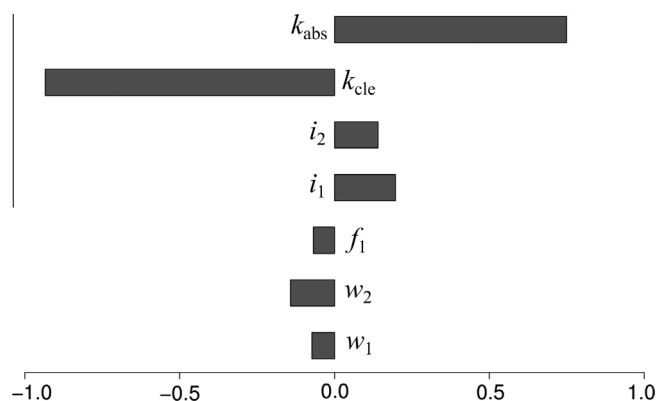


Fig. 4. Horizontal bar plot of Spearman correlation coefficients between the seven parameters w_1 , w_2 , f_1 , i_1 , i_2 , k_{abs} , k_{cle} from Table 2 and the PCDD/F concentration of the pig at slaughter age (220 d).

to the five parameters of the growth and feed functions (w_1, w_2, f_1, i_1, i_2) is small compared to the sensitivity to k_{cle} and k_{abs} . Among these five parameters, the parameters of the feed function (i_1, i_2) have the highest impact on the outcome.

3.4. Model fit to data from Lenk

Lenk (2007) conducted a feeding experiment with three groups of pigs receiving different concentrations of a PCDD/F-PCB mixture. The pigs were administered capsules with a PCDD/F-PCB mixture corresponding to feed levels of 1, 10 and 100 ng TEQ kg⁻¹ feed dry matter. At three time points after exposure, biopsy fat samples were taken and analysed. In analogy to Hoogenboom et al. (2004) data, we performed fits to the Lenk data (Appendix B) to estimate absorption fractions and metabolic clearance rates from Lenk's experiment. For the three concentrations, the mean values of the fit results of k_{cle} show a trend of increasing clearance rate for increasing concentration, but the differences are not significant (95% confidence intervals). Furthermore, the results are in accordance with the clearance rates from the fit of Hoogenboom et al. (2004) data. The absorption fractions calculated from Lenk's data decrease by increasing PCDD/F concentration. The absorption fractions from Hoogenboom's data are similar to those of Lenk for the concentration 100 ng TEQ kg⁻¹ feed.

3.5. Application on Spitaler et al. (2005)

In the study of Spitaler et al. (2005) fat samples were taken only at a single time point. Pigs received PCDD/F contaminated feed (including 1,2,3,4,7,8-HxCDD) of different levels. After slaughtering, fat samples from three different fat tissues were analysed. We applied our probabilistic model on their feed concentrations. The calculated curves did not describe the experimental 1,2,3,4,7,8-HxCDD concentrations in pig fat. Additionally, the measured concentrations in different tissues are different, in contrast to previous experience and our assumption of equal distribution.

4. Discussion

Here, a probabilistic model was combined with a kinetic model to describe, for the first time, the concentration of PCDD/F in growing pigs in statistical terms. The kinetic model was calibrated with experimental values of Hoogenboom et al. (2004). The Levenberg–Marquardt algorithm (Marquardt, 1963) was used to estimate absorption fraction and metabolic clearance rate of the PCDD congenere 1,2,3,4,7,8-HxCDD. With these fraction, rate and coefficients for growth- and feed functions (Pfeiffer et al., 1984;

Staudacher, 2010), the probabilistic model was able to reproduce and describe the German dioxin incident of winter 2010/2011 (Abraham et al., 2011) and the German PCDD/F background contamination (Schwind et al., 2009). The new probabilistic model was also compared with earlier modeling approaches. The results of these earlier, non-probabilistic approaches can be reproduced with our model.

Fries (1996) implemented a prediction model for concentrations of lipophilic chemicals in growing pigs. This model was composed of a single fat compartment and also required growth and feed functions (Doornenbal, 1971, 1972, 1975; Baile et al., 1987). Fries performed worst-case calculations with complete absorption and without metabolic clearance and excretion; uncertainties and correlations were not included. We performed calculations with k_{abs} , k_{cle} , k_{exc} , C_l restricted to the values of Fries (1996). The five parameters of growth and feed functions (Table 2) were sampled as before. At slaughter age, the mean value of the restricted probability model matches the value of Fries (Appendix B).

Hoogenboom et al. (2007) simulated PCDD/F concentrations in pig fat with a toxicokinetic two compartment model for growing pigs, without probabilistics. In spite of the differences, the resulting mean value curves of our model (Appendix B) are very similar to the curves in Hoogenboom et al. (2007). The increase and decrease of Hoogenboom et al. (2007) curves is slightly faster. That might be due to the different dynamics of their two compartment model and to the different metabolic clearance rates and absorption fractions of that congener mixture (we used the parameters which were fitted for 1,2,3,4,7,8-HxCDD, while they used the rates that were fitted to the total TEQ values).

Shen et al. (2012) repeated the feeding experiment of Lenk (2007). Moreover, they fitted exponential functions to the PCDD/F concentrations and to the absolute PCDD/F amounts of the pigs. By this procedure they obtained two rates. The first is an overall rate for dilution, metabolic clearance and excretion. The second is an overall rate for metabolic clearance and excretion. This approach has some drawbacks: It neglects the metabolic clearance that takes place during the time of contaminated feeding and it neglects excretion. Finally, a simple exponential function is only a rough approximation for the solution of the differential equation Eq. (A.4). Hence, it is not surprising that our metabolic clearance rate is a factor of three smaller than the rate for metabolic clearance and excretion of Shen et al. (2012). Apart from that, the predicting equation they present does not take into account any uncertainty of the rates or spreading of the feed contamination or interval.

The metabolic clearance rate from the fit of Hoogenboom et al. (2004) data has a relative error of 42%. That might be surprising, because the experimental values seem to follow an exponential decay without noteworthy deviations. But the main part of the concentration decay is due to dilution by growth. The metabolic clearance is an additional process and changes the overall shape only slightly. Only large values of k_{cle} would change the shape of the concentration curve. Therefore, the large 95%-error interval for the half life of [100,1040] d with the estimated mean of $t_h = 180$ d is reasonable. From Lenk (2007) data, we obtain relative errors between 40% and 100%.

The absorption fraction of 46% from Hoogenboom's data matches the data given for rats fed with TCDD contaminated diet (Fries and Marrow, 1975). The model fit to the data of Lenk results with higher absorption fractions. This might result from the different matrix of administration. While in Hoogenboom's experiment, the contaminated feed of the Belgian incident was used, Lenk used a PCDD/F mixture, that was dissolved in oil and administered in capsules. From these different application forms, one can expect different absorption fractions. In accordance with (Canady et al., 2002), absorbed fractions are higher for the Lenk experiment.

The sensitivity analysis suggests that the outcome is most sensitive to clearance rate and absorption fraction. But although we might not know the clearance rate and absorption fraction of congeners in a future dioxin incident, the model still can produce valuable information. This is because under realistic conditions, the influence of feed contamination and interval of contaminated feeding has more influence on the outcome than the parameters of the sensitivity analysis. Beyond that, it would be useful to have experimental data for all PCDD/F congeners. Even though the metabolic clearance rate is not as important as expected for the probabilistic model, it would improve the reliability of the model.

4.1. Conclusion

The approach presented here can be a useful decision tool during dioxin incidents. It can be used to estimate whether or not a marketable food can result after a period of clean feed following a period of contaminated feed. This can avoid either unnecessary culling of animals or financial damages to farmers who feed pigs until slaughter age although no marketable product will result from them.

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Appendix A. Differential equation for the PCDD/F concentration

The change of PCDD/F amounts in the body of a pig is the sum of incoming and outgoing PCDD/F amounts, see Eq. (1). These in- and outgoing PCDD/F amounts are:

$$\begin{aligned} A_{\text{intake}} &= A_I(t) k_{\text{abs}}, & A_{\text{clearance}} &= A(t) k_{\text{cle}}, \\ A_{\text{excretion}} &= C(t) k_{\text{exc}} \end{aligned} \quad (\text{A.1})$$

The amount of PCDD/F in the feed $A_I(t)$ equals the product of feed amount and concentration in feed:

$$A_I(t) = I(t) C_I(t) \quad (\text{A.2})$$

Summation of the terms in Eq. (A.1), considering Eq. (A.2), yields Eq. (1). The PCDD/F concentration in the body fat of the pig equals the PCDD/F amount divided by the weight of body fat:

$$C(t) = A(t)/F(t) \quad (\text{A.3})$$

Introducing (A.3) into Eq. (1) and considering the chain rule, yields a differential equation for the PCDD/F concentration in body fat (A.4).

$$\frac{d}{dt} C(t) = \frac{1}{F(t)} \left\{ I(t) C_I(t) k_{\text{abs}} - C(t) \left[\frac{d}{dt} F(t) + F(t) k_{\text{cle}} + k_{\text{exc}} \right] \right\} \quad (\text{A.4})$$

With the growth- and feed functions (Eqs. (2)–(4)), one gets a differential equation for $C(t)$, Eq. (A.5), containing 9 parameters ($w_0, w_1, w_2, f_1, i_1, i_2, k_{\text{exc}}, k_{\text{abs}}, k_{\text{cle}}$). To get the concentration as a function of time, one has to integrate Eq. (A.5) numerically.

$$\begin{aligned} \frac{d}{dt} C(t) &= \frac{1}{f_1 t (w_0 + w_1 t^{w_2})} \{ i_1 [1 - e^{-i_2 (w_0 + w_1 t^{w_2})}] \\ &\quad C_I(t) k_{\text{abs}} - C(t) [f_1 (w_0 + w_1 (1 + w_2) t^{w_2}) \\ &\quad + f_1 t (w_0 + w_1 t^{w_2}) k_{\text{cle}} + k_{\text{exc}}] \} \end{aligned} \quad (\text{A.5})$$

Appendix B. Supplementary material

Supplementary content to this article can be found online at: <http://dx.doi.org/10.1016/j.chemosphere.2013.06.015>.

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